***Diffie-Hellman Key Exchange Protocol***

The Diffie-Hellman key exchange protocol establishes a secret key between two parties and only involves the exchange of public (non-encrypted) information to do so. The original protocol involves a prime modulus $p$ and a base $g$, where $g$ is a primitive root of $p$ (i.e. $g^k \bmod p$ can be used to generate all $p-1$ integers relatively prime to $p$.

Both $p$ and $g$ are published.

Person A chooses a secret power $a$ and sends $g^a \bmod p$ to person B.

Person B chooses a secret power $b$ and sends $g^b \bmod p$ to person A.

Person A computes $s = (g^b \bmod p)^a \bmod p$

Person B computes $s = (g^a \bmod p)^b \bmod p$

Person A and B both share a secret encryption key $s = g^{ab} \bmod p$

**Example:**

$p$=991 and $g$=6 are published.

A chooses a secret power $a$=123 and sends $6^{123} \bmod 991 = 541$ to B.

B chooses a secret power $b$=456 and sends $6^{456} \bmod 991 = 644$ to A.

A computes: $s = (644)^{123} \bmod 991 = 265$.

B computes: $s = (541)^{456} \bmod 991 = 265$.

Person A and B both share a secret encryption key $s = 265 = 6^{(123)(456)} \bmod 991$

Note that with the published values of $p$=991 and $g$=6 and the non-encrypted messages 541 and 644 we could find the secret key $s$ by checking all values of $a$ for $6^a \bmod 991 = 541$ and all values of $b$ for $6^b \bmod 991 = 644$.

Using a <u>large prime modulus</u> with around 300 digits, and secret powers $a$ and $b$ with around 100 digits makes this computationally unfeasible.

**Exercises:**

1. Two people (A and B) establish a shared secret key using the Diffie-Hellman key exchange protocol. They use the modulus of $p$=103 and a base $g$=5. Person A chooses a secret power $a$=16. Person B chooses a secret power $b$=8.
   a) What number does A send B?
   b) What number does B send A?
   c) What is the shared secret key?

2. *Andross* and *Bowser* establish a shared secret boss key using the Diffie-Hellman key exchange protocol. They use the modulus of $p$=17 and a base $g$=3. *Andross* sends *Bowser* the number 15. *Bowser* sends *Andross* the number 4.
   a) What are the secret powers ($a$ and $b$) of *Andross* and *Bowser*?
   b) What is the shared secret boss key?

**Answers:**

**1.**

a) 32
b) 49
c) 60

**2.**

a) $a$=6, $b$=12
b) $(10000)_2$